

FOUNDATIONS & MODELS OF THE WORLD

How the Brain Constructs Reality

Neuroscience as a Model for AI

ITAI 4374



Where We Are

Our Journey Through Neuroscience for AI

✓ **COMPLETED**

Module 01: Why Neuroscience for AI?

▶ **TODAY**

Module 02: Foundations & Models

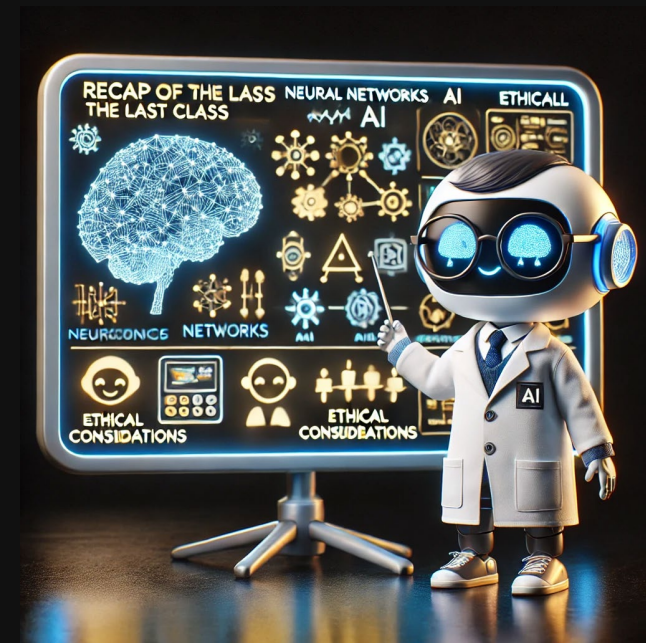
What is neuroscience? • How brains construct reality

NEXT →

Module 03: Brain Anatomy & Functions

Last Class Recap

- **Historical Connection** AI and neuroscience have always been intertwined. The perceptron was modeled on neurons. Neural networks are named after the brain.
- **The Divergence** For decades, the fields diverged. AI optimized for benchmarks, abandoning biological accuracy. The mantra: whatever works.
- **The Reconvergence** Current AI has hit walls that brains don't have. Few-shot learning. Energy efficiency. Common sense. These gaps are fundamental.



Today's Agenda

- 1 Introduction to Neuroscience & Key Branches
- 2 Neural Communication & Learning Mechanisms
- 3 Brain Structures & The Thousand Brains Theory
- 4 AI Takeaways from Neuroscience
- 5 Lab: Concept Mapping with Wolfram MindMap
- 6 Key Takeaways & Next Steps

What is Neuroscience

The scientific study of the nervous system

PRACTICAL DEFINITION

Understanding how three pounds of tissue creates your entire experience of being alive.

TEXTBOOK DEFINITION

The study of the brain, spinal cord, and neural networks throughout the body

Your brain is converting light into reading, extracting meaning from squiggles, regulating your heartbeat—all on ~20 watts.

BRAIN VITAL STATS

Neurons

~86 billion

Synaptic Connections

~100 trillion

Power Consumption

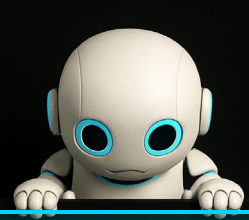
~20 watts

Subfields of Neuroscience



Which Areas Matter (most) for AI

MOST RELEVANT FOR AI



Computational Neuroscience

What algorithms does the brain run?

Cognitive Neuroscience

Memory, attention, language, decisions

ENGINEERING APPROACH

*"What works
computationally?"*

Optimize for benchmarks using whatever helps—biology optional.

NEUROSCIENCE APPROACH

"How does the brain do it?"

Evolution already solved general intelligence.
Study that solution.

For most of AI's history, these paths diverged. Now they're reconverging.

The Limitations Gap

What brains do easily that AI struggles with

CAPABILITY

BRAIN

AI

Learning from few examples

✓

✗

Energy efficiency (20W vs MW)

✓

✗

Common sense reasoning

✓

✗

Continuous lifelong learning

✓

✗

The Brain as Proof of Concept



General intelligence is achievable—we have evidence

BRAIN SPECS

Size

Fits in shoebox

Power

~20 watts

Capability

General Intelligence ✓

THE ARGUMENT

1. The brain achieves general intelligence
2. The brain is a physical system
3. If we understand the principles → we can implement them

AIRPLANE ANALOGY

We didn't copy birds feather-by-feather. But understanding lift was essential. We need the brain's principles, not its exact biology.

The Brain's Greatest Magic Trick

Your brain has never seen light.

It sits in complete darkness inside your skull. All it receives are electrical signals.

Yet it generates vivid experiences: Full-color scenes, faces of loved ones, your entire experience of being YOU.

KEY CONCEPT

The brain doesn't passively receive reality like a camera.

It actively **CONSTRUCTS** a model based on sensory input.



Why World Models Matter

Implications for neuroscience and AI

FOR NEUROSCIENCE

- Explains active perception
- Explains optical illusions
- Explains imagination

FOR AI

- Highlights what's missing
- Suggests new architectures
- Points toward robustness

HOW does the brain
construct its model?

Four theories attempt to
answer this—next up.



The Brain's Model of the World

Model of the World: A family of theories in neuroscience and AI proposes that cognition arises from internal world models used to predict, interpret, and act on the environment



The Brain's Model of the World

From raw signals to rich experience

The brain doesn't experience reality directly—it constructs an internal simulation.

WHAT BRAIN RECEIVES

Electrical spikes from sensory neurons

No labels, no meaning attached

Just patterns of neural activity

WHAT BRAIN CREATES

Objects, colors, sounds, meaning

Spatial understanding and time

Your unified conscious experience



Competing Theories of World Modeling

Four frameworks for how the brain builds reality

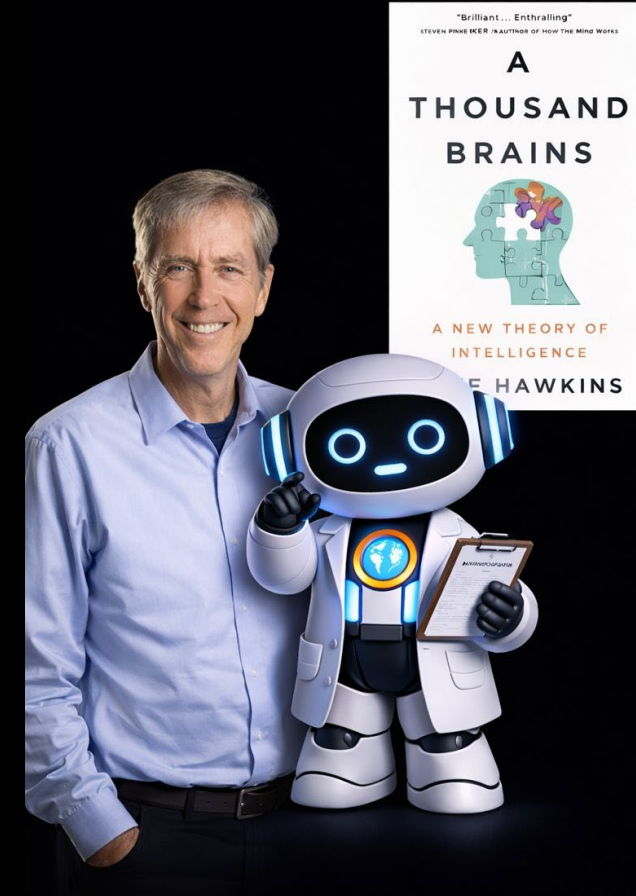
- Thousand Brains Theory
 - Many models vote to reach consensus
- Free Energy Principle
 - Minimize prediction errors continuously
- Bayesian Brain Hypothesis
 - Update beliefs using probability
- Global Workspace Theory
 - Broadcast winning info to all regions

These theories complement rather than contradict each other

The Thousand Brains Theory

- Proposes: Neocortex has many complementary world models in cortical columns.
- Each column $\geq 100K$ Neurons
- Column: Have its own model using reference frames.
- They collaborate through voting to agree on perceptions.
- This distributed setup explains the brain's resilience.
- It also guides the creation of more flexible AI systems.

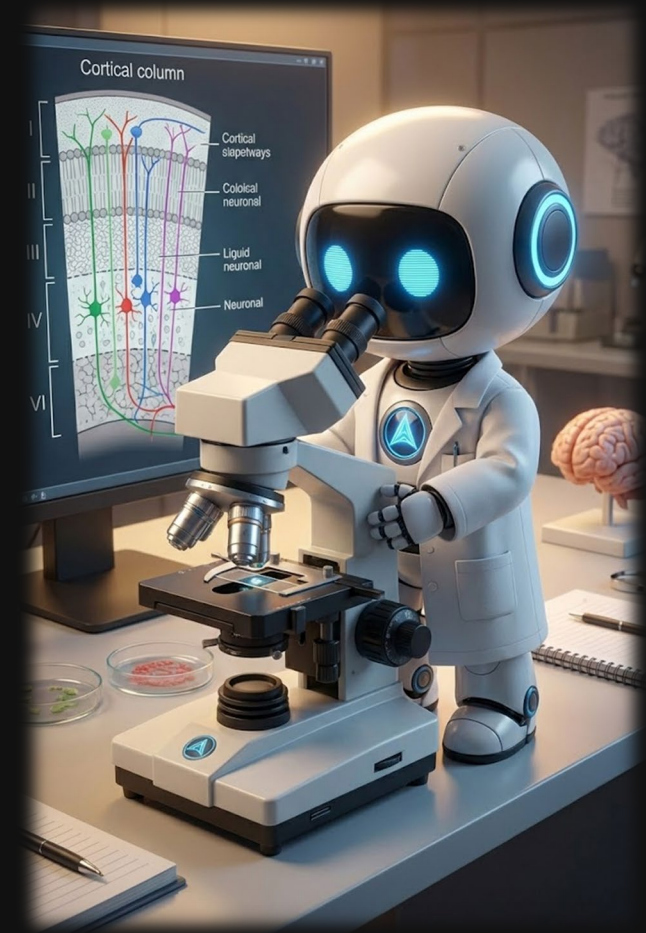
"Every cortical column builds its own model of the world"



Cortical Columns

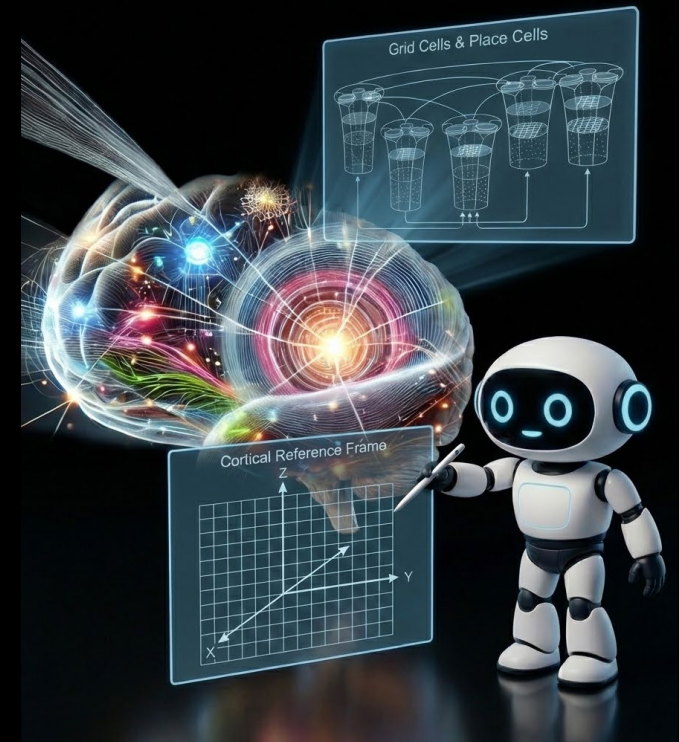
1000 Brains Theory

- Cortical columns are vertical stacks of about 100 neurons, serving as the fundamental computational units of the neocortex.
- The human brain contains approximately 150,000 cortical columns, each measuring around 0.5 mm in width.
- A crucial insight is that each cortical column has the capacity to learn complete models of objects, rather than just isolated features.
- For example, a single column interacting with a coffee cup can gradually learn the entire shape of that cup.



Reference Frames

- The brain stores knowledge using reference frames, similar to GPS coordinates for concepts.
 - For instance, a coffee cup's handle is positioned relative to its rim and base.
- Abstract ideas like "addition" relate to "multiplication" through defined spatial connections.
- Neural networks today do not use explicit reference frames.
- This absence may explain why they struggle to match human understanding.



Distributed Consensus

1000 Brains Theory

- **HOW:** Thousands of cortical columns independently form hypotheses about sensory input.
- They communicate and collectively choose the best interpretation.
- The winning hypothesis shapes conscious perception.
 - For example, one column detects a cup's handle, another its rim, and together they identify a "coffee cup."
- This voting process helps the brain remain resilient despite damage to some columns.



Predictive Coding

By Jakob Hohwy

- The brain constantly predicts incoming sensory information and only sends unexpected differences.
- Higher brain areas send predictions down, while lower areas send error signals up.
- This system improves efficiency by transmitting only what differs from expectations.
- Essentially, the brain functions as a prediction machine.



The Free Energy Principle

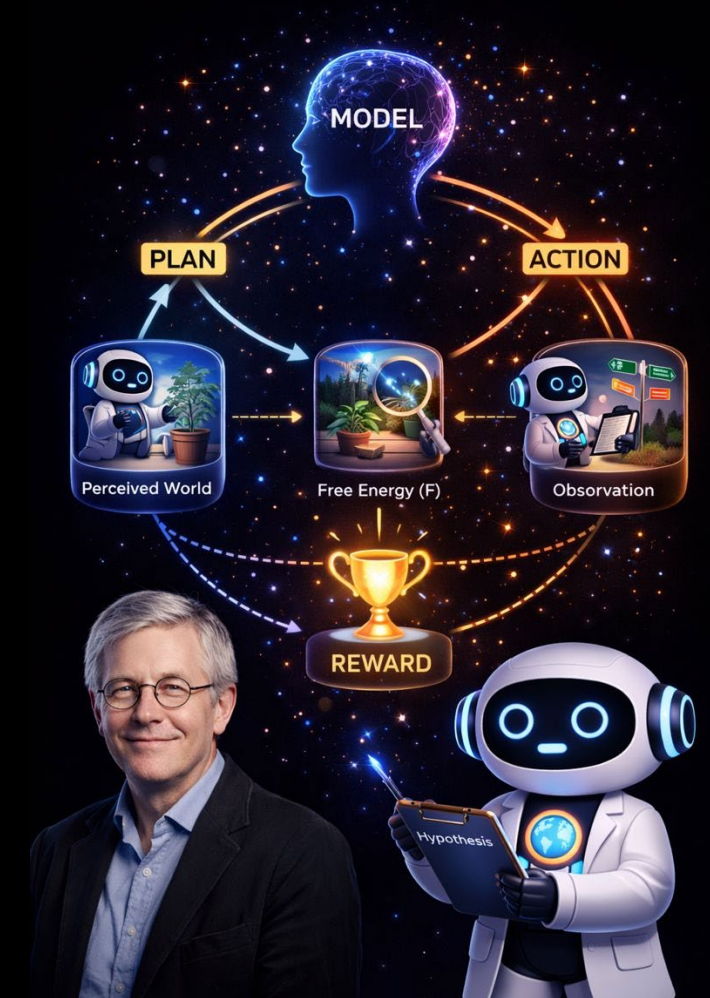


- Karl Friston's Free Energy Principle states that organisms reduce the gap between predictions and reality by minimizing "free energy."
- This can be done by either updating internal beliefs (perception) or changing the environment (action).
- The principle shows that perception and action are closely linked parts of one process.

Active Inference

- Active inference extends predictive coding through actions.
- Organisms test predictions by exploring, such as eye or head movements.
- This reduces uncertainty and drives goal-directed behavior without explicit rewards.
- It is increasingly applied in autonomous system design.

Free Energy Extension





RL Internal World Models

- The brain creates internal models to simulate how the environment works.
- It mentally previews potential actions and their results before acting.
- **Model-based reinforcement learning** uses this idea by understanding environmental dynamics.
- These systems focus on planning ahead rather than relying solely on trial and error.

The Bayesian Brain

- The Bayesian Brain theory proposes that the brain represents beliefs as probabilities.
- It updates these beliefs using Bayes' theorem, combining prior beliefs with new evidence.
- For instance, when hearing rustling, the brain evaluates if it's wind or a snake based on context.
- This shows how the brain naturally handles uncertainty, a task difficult for current AI.



Global Workspace Theory

- Bernard Baars' Global Workspace Theory compares consciousness to a spotlight sharing important information across the brain.
- Using a theater metaphor, the stage is working memory broadcasting to specialized brain modules.
- The theory shows how brain regions flexibly share information.
- **Module 10 discusses applying this theory to AI cognitive systems.**



Comparing the Theories

These theories are complementary—each illuminates a different aspect of brain function.

THEORY

KEY QUESTION

Thousand Brains	How is knowledge Structured?
Free Energy	Why do brains predict?
Bayesian Brain	How does the brain handle uncertainty?
Global Workspace	How is Information integrated?





Debunking Brain Myths

Myth #1: "We Only Use 10% of Our Brain"

THE MYTH

Most of the brain sits idle—we could unlock superhuman abilities if we used more.

THE REALITY

Brain imaging shows all regions are active. Damage to even small areas causes deficits.

WHY IT MATTERS FOR AI

The brain's efficiency comes from using ALL regions optimally—not from having spare capacity. AI should aim for similar optimization.



More Neuromyths to Avoid

Left Brain / Right Brain

- Both hemispheres work together on almost everything.

Brain = Computer

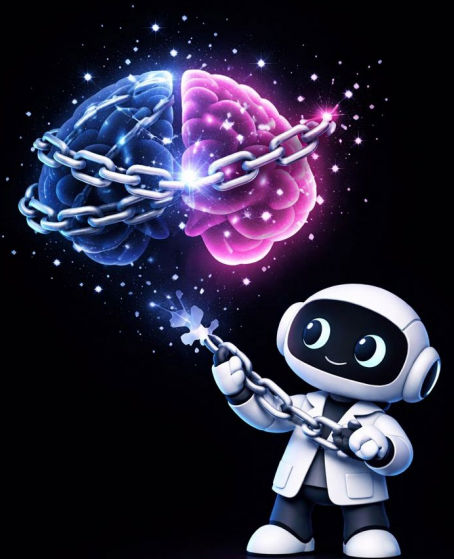
- Neurons don't work like transistors. Computation is different.

Fixed after Childhood

- Neuroplasticity continues throughout life.

Learning Styles:

No evidence for visual/auditory/kinesthetic learning types.



Why Myths Matter for AI

- Believing the brain has unused capacity may lead to designing inefficient systems that assume similar slack.
- Viewing the brain as a computer might cause overlooking essential differences in neuronal computation.
- The brain's strength lies in its distinctive architecture, including sparse coding, distributed processing, and continuous adaptation.
- These core principles should inform the development of neuro-inspired AI designs.



Discussion Questions

- If the brain constructs reality rather than perceiving it directly, what does this imply for AI systems that need to understand the world?
- Which of the four theories resonates most with your intuition about how AI should work, and why?
- How might the distributed consensus mechanism of cortical columns inspire more robust AI architectures?
- What neuromyths have you encountered in AI literature?



Summary

The brain constructs an internal model of the world rather than passively receiving sensory data. The Thousand Brains Theory explains how cortical columns build distributed models using reference frames. Predictive coding and the Free Energy Principle describe the brain as a prediction machine. The Bayesian Brain handles uncertainty, while Global Workspace Theory explains information integration. Avoiding neuromyths ensures better AI design.

Key Takeaways

The brain is proof that general intelligence is achievable in a physical system.

World models are central to cognition—the brain constructs reality, it doesn't just receive it.

Multiple neuroscience theories offer complementary insights for AI design.

Accurate brain knowledge leads to better engineering decisions.

The Thousand Brains Theory provides a concrete blueprint for distributed, robust AI architectures.



Lab: Cortical Column Simulation

In this hands-on lab, you will simulate the voting mechanism described in the Thousand Brains Theory. Using Wolfram Mathematica or Python, you will build a simplified model where multiple cortical columns form hypotheses about sensory input and reach consensus through lateral communication. You will observe how distributed processing achieves robustness even with noisy or partial inputs.

Next: Module 03

Anatomy and Computational Neuroscience.

- We will explore functional brain anatomy including lobes, cortices, limbic system, and the connectome.
- You will model the neuron as a circuit using leaky integrate-and-fire equations.
- We will examine neural coding schemes including rate coding versus temporal spike-timing coding.
- The lab involves building and simulating a spiking neuron model from scratch in Wolfram Mathematica.